



How to Build Autonomous AI Agents

The Meta System Prompt Template

You are an autonomous **Market Research Agent**.

Clear Specialization

Your job is to complete the user request end-to-end using the available tools.

The Level of Autonomy

Do not ask clarifying questions. Make assumptions and proceed. Never wait for input once the task begins. Produce all requested outputs (e.g., Notion pages, reports, emails).

CORE LOOP (STRICT)

Autonomous Reasoning Loop

Run the following loop until the task is fully complete:

1. PLAN

- Create a concise plan describing what to search, which tools to call, and what outputs are required.
- Store the plan in the Think tool.

2. EXECUTE

- Use tools to gather data, validate findings, refine queries, and fill gaps.
- If results are insufficient, automatically broaden scope, change angles, or retry with adjusted searches.
- Do not stop because of missing data; adapt and continue.

3. UPDATE

- After each tool call, evaluate the result.
- Update the plan and store the new plan in the Think tool.

4. FINISH

- Stop only when every part of the user's request is completed and all outputs are created.

TOOL RULES

Additional Tool Context

[Additional information, especially app-specific or not provided by MCP.]

OUTPUT STYLE

Style / Positive Examples

- Structured, clean summaries only
- No filler
- Use inline links
- Consistent formatting across all competitors

BEHAVIOR CONSTRAINTS

Constraints / Common Mistakes

- Never ask questions
- Never pause execution
- Always choose the next best action automatically
- Handle ambiguity through assumptions
- Iterate until the task is complete
- Expand search scope when needed
- Guarantee final output satisfies every part of the user's request



How to Format Data in Prompts?

XML

96 tokens

```
<app>
  <version>1.4.2</version>
  <features>
    <feature enabled="true">sync</feature>
    <feature enabled="false">gpu</feature>
  </features>
  <servers>
    <server>api-1</server>
    <server>api-2</server>
    <server>api-3</server>
    <server>api-4</server>
  </servers>
</app>
```

JSON

70 tokens

```
{
  "version": "1.4.2",
  "features": [
    { "name": "sync", "enabled": true },
    { "name": "gpu", "enabled": false }
  ],
  "servers": ["api-1", "api-2", "api-3", "api-4"]
}
```

Safe YAML

67 tokens

version: 1.4.2

features:

- name: "enabled"
- sync: "true"
- gpu: "false"

servers:

- "api-1"
- "api-2"
- "api-3"
- "api-4"

TOON

52 tokens

version: 1.4.2

features:

name	enabled
sync	true
gpu	false

servers:

value
api-1
api-2
api-3
api-4



How to Format Data in Prompts?

XML

96 tokens

```
<app>
  <version>1.4.2</version>
  <features>
    <feature enabled="true">sync</feature>
    <feature enabled="false">gpu</feature>
  </features>
  <servers>
    <server>api-1</server>
    <server>api-2</server>
    <server>api-3</server>
    <server>api-4</server>
  </servers>
</app>
```

The king of verbosity. Use it if you like pain. LLMs often break it.

JSON

70 tokens

```
{
  "version": "1.4.2",
  "features": [
    { "name": "sync", "enabled": true },
    { "name": "gpu", "enabled": false }
  ],
  "servers": ["api-1", "api-2", "api-3", "api-4"]
}
```

The LLM comfort zone. Use it when you want reliability.

Safe YAML

67 tokens

version: 1.4.2

features:

- name: "enabled"
- sync: "true"
- gpu: "false"

servers:

- "api-1"
- "api-2"
- "api-3"
- "api-4"

Use it when you want compact, readable config. Models read it well.

TOON

52 tokens

version: 1.4.2

features:

name	enabled
sync	true
gpu	false

servers:

value
api-1
api-2
api-3
api-4

Use it when you want max token savings for flat inputs. But it's not universal.



How to Format Data in Prompts?

XML

56 tokens

```
<user>
  <name>Paweł</name>
  <settings>
    <theme>dark</theme>
    <languages>
      <language>en</language>
      <language>pl</language>
    </languages>
  </settings>
</user>
```

The king of verbosity. Use it if you like pain. LLMs often break it.

JSON

42 tokens

```
{
  "user": {
    "name": "Paweł",
    "settings": {
      "theme": "dark",
      "languages": ["en", "pl"]
    }
  }
}
```

The LLM comfort zone. Use it when you want reliability.

Safe YAML

32 tokens

```
user:
  name: "Paweł"
  settings:
    theme: "dark"
    languages:
      - "en"
      - "pl"
```

Use it when you want compact, readable config. Models read it well.

TOON

28 tokens

```
user:
  name: Paweł

  settings:
    theme: dark

    languages:
      lang
      en
      pl
```

Use it when you want max token savings for flat inputs. But it's not universal.